



IILM UNIVERSITY

School of Computer Science & Engineering

INTERNSHIP PRESENTATION

DATA ANALYTICS INTERNSHIP

Organization: CodeAlpha

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Enrollment No.: Engineering/2024/13627

Roll No.: 2410030180

Program: B.Tech Computer Science & Engineering

Specialization: Data Science

Semester: 5th Semester

Session: 2024–2028



ABOUT THE INTERNSHIP

Internship Domain: Data Analytics

Organization: CodeAlpha

Tools / Environment: Jupyter Notebook

Duration: 1 Month

Internship Overview

During my internship with **CodeAlpha**, I worked on practical Data Analytics tasks using Python. The internship gave me an opportunity to apply concepts such as **data collection, data cleaning, exploratory analysis, and visualization** to real datasets.

I worked on three main tasks:

01 — Web Scraping

Collected book information from the *Books to Scrape* website using Python.

02 — Exploratory Data Analysis

Worked with an **Amazon Sales dataset downloaded from Kaggle** and explored its structure and quality.

03 — Data Visualization

Created visualizations to understand product categories, pricing, ratings, and discounts.

My Learning Approach



Collect → Clean → Explore → Visualize → Understand

This internship helped me understand how the different stages of a basic **Data Analytics workflow** connect with each other.

Internship Objectives

During this internship, I wanted to understand how Data Analytics is actually performed on practical datasets and improve my confidence in working with Python.

My main objectives were:

• To improve my Python programming and data-handling skills.	• To create meaningful visualizations using Matplotlib and Seaborn.
• To learn how to collect and work with real-world data.	• To learn the basics of Web Scraping using Requests and BeautifulSoup.
• To understand the process of data cleaning and preprocessing.	• To become more comfortable working with Jupyter Notebook and Kaggle datasets.
• To practice Exploratory Data Analysis (EDA) and understand dataset patterns.	• To develop better problem-solving and analytical skills through hands-on tasks.

My Goal:

To move from understanding Data Analytics concepts theoretically to actually applying them in practical projects.

TOOLS & TECHNOLOGIES

During the internship, I used different tools and Python libraries depending on the task.

Tool	Purpose
 Python	Main programming language used for the projects
 Jupyter Notebook	Writing, running, and testing code
 Pandas	Data loading, cleaning, manipulation, and analysis
 NumPy	Numerical operations and data handling
 Matplotlib	Creating and customizing basic charts
 Seaborn	Creating statistical and informative visualizations
 Requests	Sending HTTP requests to retrieve webpage content
 BeautifulSoup	Parsing HTML and extracting information from webpages
 Kaggle	Source of the Amazon sales dataset

HOW I USED THEM



Together, these tools helped me follow the basic Data Analytics workflow from collecting data to analyzing and visualizing it.

Task 1: Web Scraping

Objective

The main objective of this task was to understand how data can be collected automatically from a website and converted into a structured format that can be used for further analysis.

Website Used Books to Scrape: <https://books.toscrape.com/>

I selected this website because it provides a simple environment for practicing web scraping.

Tools Used

Python • Requests • BeautifulSoup • Pandas • Jupyter Notebook

Process

01 — Send Request: Used `requests.get()` to access the webpage.

02 — Get HTML Content: Retrieved the webpage's HTML content.

03 — Parse HTML: Used BeautifulSoup to understand the webpage structure.

04 — Extract Data: Located the book elements & extracted their titles.

05 — Store Data: Stored the extracted titles in a Python list.

06 — Create DataFrame: Converted the list into a Pandas DataFrame.

07 — Export Dataset: Saved the final data as `scraped_books.csv`

Result: Successfully collected book titles from the website and stored them in a structured CSV file.

[1]

✓ 0s

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
```

[2]

✓ 0s

```
url = "https://books.toscrape.com/"

response = requests.get(url)

soup = BeautifulSoup(response.text, "html.parser")
```

[3]

✓ 0s

```
books = soup.find_all("h3")

book_names = []

for book in books:
    book_names.append(book.a["title"])

book_names[:10]
```

▼

```
... ['A Light in the Attic',
     'Tipping the Velvet',
     'Soumission',
     'Sharp Objects',
     'Sapiens: A Brief History of Humankind',
     'The Requiem Red',
     'The Dirty Little Secrets of Getting Your Dream Job',
     'The Coming Woman: A Novel Based on the Life of the Infamous Feminist, Victoria Woodhull',
     'The Boys in the Boat: Nine Americans and Their Epic Quest for Gold at the 1936 Berlin Olympics',
     'The Black Maria']
```

[4]

✓ 0s

```
scraped_data = pd.DataFrame({
    "Book Name": book_names
})

scraped_data.head()
```

▼

Index		Book Name
0	A Light in the Attic	
1	Tipping the Velvet	
2	Soumission	
3	Sharp Objects	
4	Sapiens: A Brief History of Humankind	

Show 25 per page
Like what you see? Visit the [data table notebook](#) to learn more about interactive tables.

[6]

✓ 0s

```
scraped_data.to_csv("scraped_books.csv", index=False)
```

Result: Book titles were successfully extracted from the webpage and saved as scraped_books.csv

TASK 2 — EXPLORATORY DATA ANALYSIS

Objective

To understand the structure and quality of the dataset before performing further analysis and visualization.

DATASET
Amazon Sales Dataset
Source: Kaggle
Environment: Jupyter Notebook
Format: CSV
 <pre>df = pd.read_csv("amazon.csv.zip") df.head()</pre>

What I Did:

EDA PROCESS
01 - Imported the Dataset Loaded the CSV file into Jupyter Notebook using Pandas.
02 - Inspected the Data Used head(), shape, info(), and columns to understand the dataset.
03 - Checked Data Quality Checked for:Missing values Duplicate records Unnecessary spaces in column names Data types
04 - Cleaned the Data Removed duplicate records and cleaned column names where required.
05 - Performed Basic Analysis Used describe() and other Pandas functions to understand the numerical data and overall dataset.

Result:The dataset was cleaned and explored, making it ready for the next stage — **Data Visualization**.

SAMPLE CODE

```
df.describe()
```

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_pro
count	1465	1465	1465	1465	1465	1465	1465	1463	
unique	1351	1337	211	550	449	92	28	1143	
top	B083342NKG	Fire-Bolt Ninja Cat Pro Plus 1.83" Smart Wat...	Computers&Accessories(Accessories&Peripherals)...	₹199	₹999	50%	4.1	9,378	[CHARACTERS] FUNCTIONS This comes
freq	3	5	233	53	120	56	244	9	

```
df.isnull().sum()
```

	0
product_id	0
product_name	0
category	0
discounted_price	0
actual_price	0
discount_percentage	0
rating	0
rating_count	2
about_product	0
user_id	0
user_name	0
review_id	0
review_title	0
review_content	0
img_link	0
product_link	0

check dataset size

df.shape

(1465, 16)

```
df.duplicated().sum()
df.drop_duplicates(inplace=True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1465 entries, 0 to 1464
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   product_id            1465 non-null   object
1   product_name          1465 non-null   object
2   category              1465 non-null   object
3   discounted_price      1465 non-null   object
4   actual_price          1465 non-null   object
5   discount_percentage   1465 non-null   object
6   rating                1465 non-null   object
7   rating_count          1463 non-null   object
8   about_product         1465 non-null   object
9   user_id               1465 non-null   object
10  user_name             1465 non-null   object
11  review_id             1465 non-null   object
12  review_title          1465 non-null   object
13  review_content        1465 non-null   object
14  img_link              1465 non-null   object
15  product_link          1465 non-null   object
dtypes: object(16)
memory usage: 183.3+ KB
```


TASK 3 — DATA VISUALIZATION

Turning data into clear and meaningful insights

OBJECTIVE

The objective of this task was to represent the Amazon sales data visually and make it easier to identify patterns and compare different aspects of the products.

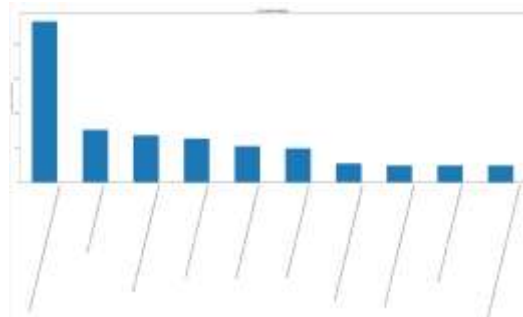
Tools Used:

• Matplotlib • Seaborn • Pandas

VISUALIZATIONS

01 - Top 10 Product Categories

Bar Chart



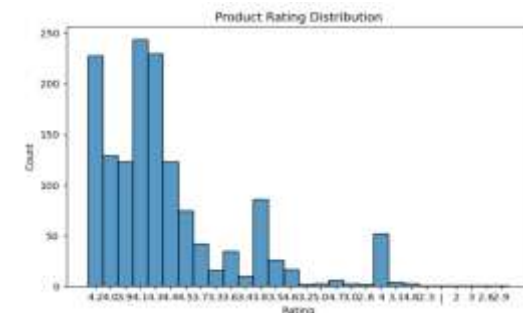
02 - Actual Price vs. Discounted Price

Regression Plot



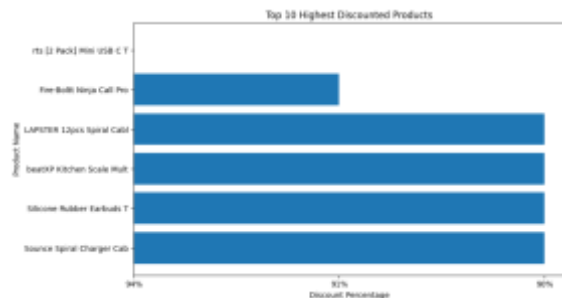
03 - Product Rating Distribution

Histogram



04 - Top 10 Highest Discounted Products

Horizontal Bar Chart



Visualization helped me understand the dataset more quickly than looking at raw rows of data.

KEY OBSERVATIONS

- Product categories vary in the number of products listed.
- Actual price and discounted price show a clear relationship.
- Product ratings are generally concentrated toward the higher rating range.
- The discount chart highlights products with the highest discount percentages.

LEARNING OUTCOMES

What I learned during the internship

Python & Jupyter Improved my confidence in writing, running and understanding python code.	Data Cleaning Learning how to identify duplicates, missing values, data types, and other data issues.
Exploratory Data Analysis Learned how to explore a dataset and understand its structure and basic patterns.	Data Visualization Gained practical experience creating charts using Matplotlib and Seaborn.
Web Scraping Learned how to collect useful information from webpages using Requests and BeautifulSoup.	Problem-Solving Improved my ability to debug code, understand errors, and find practical solutions.

My biggest learning was understanding the complete basic Data Analytics workflow:



CONCLUSION

This internship gave me an opportunity to apply my classroom knowledge to practical Data Analytics tasks. I gained practical experience with **Python, Pandas, NumPy, Matplotlib, Seaborn, Requests, and BeautifulSoup**.

The experience helped me improve my **data-handling, problem-solving, and analytical skills**, and gave me a better understanding of how a basic Data Analytics workflow is carried out.

During the internship, I worked on:

Web Scraping → Exploratory Data Analysis → Data Visualization

This internship helped me turn theoretical knowledge into practical experience.

-----THANK YOU-----